

# CLIMB: Feasibility-Gated Motion Tracking for Generalist Humanoid Controllers

Anonymous Authors

**Abstract**—Outcome-adaptive curricula treat persistent tracking error as learnable difficulty. This assumption fails when a retargeted reference demands contact wrench or actuator effort that the declared robot and scene cannot supply—a failure we call *reference-physics misalignment* (RPM). We formulate tracking difficulty as the staged product of reference feasibility, bank-relative support, and intrinsic motion demand. We then introduce CLIMB, a closed-loop data-to-policy framework with three interfaces: *refeas* screens final robot-space trajectories using contact-free inverse dynamics and a torque-limited contact program; the Dynamic Feasibility and Repair Pipeline (DFRP) projects qualified contacts and re-screens the result; and an exact-support curriculum assigns zero probability to rejected intervals before capped learning-progress allocation. In a three-seed Unitree G1 campaign, top-1 allocation peaked at 87–89%, and the same unsupported kneel-and-crawl reference became a recurrent attractor. On one AMASS-to-G1 pipeline, *refeas* flags 2,442 of 10,705 clips; two implementations agree on 39 of 40 decisions in a stratified same-clip panel. DFRP qualifies 22 of 26 flagged candidates in a frozen stratified panel while leaving 4 of 4 feasible controls byte-identical. Finally, adding feasibility features raises cross-policy difficulty-transfer Spearman correlation from 0.567 to 0.609 over 100 held-out clips ( $p = 0.010$  against random features). These results establish a concrete interface for separating reference admissibility from policy learning and compute allocation.

## I. INTRODUCTION

Large retargeted motion banks have made generalist humanoid tracking a practical training recipe: map human motion to a robot, train a policy over the resulting references in massively parallel simulation, and spend additional updates on motions the policy currently fails. Systems such as PHC, MaskedMimic, H2O, ExBody, BeyondMimic, and SONIC demonstrate the scale of this recipe [1], [2], [3], [4], [5], [6]. Failure, completion, or recent learning progress then appears to identify where the next policy update is most useful [5], [7], [8].

That interpretation assumes every reference defines a solvable control problem. A retargeted trajectory may instead have no admissible contact capable of supplying its demanded base wrench, or may require joint effort beyond the modeled actuator range. We call this *reference-physics misalignment* (RPM). Under RPM, policy failure conflates three distinct quantities:

$$\mathcal{D}(r; M, E, \mathcal{B}) = \mathcal{F}(r; M, E) \times \mathcal{S}(r; \mathcal{B}) \times \mathcal{I}(r). \quad (1)$$

Here  $\mathcal{F}$  is feasibility for robot  $M$  in scene  $E$ ;  $\mathcal{S}$  is scarcity relative to training bank  $\mathcal{B}$ ; and  $\mathcal{I}$  is intrinsic reference demand, including speed, contact switching, and friction. Equation (1) is a decision factorization, not an independence assumption: if  $\mathcal{F} = 0$ , the reference is routed rather

than ranked as learnable difficulty; for admitted references, support and intrinsic demand describe what the training distribution contains and what it asks the policy to learn.

We observed the consequence in a 100-motion Unitree G1 campaign. A BeyondMimic-style sampler concentrated on the same kneel-and-crawl reference in all three seeds. Campaign peak top-1 mass was 0.884, 0.870, and 0.893; another clip owns the maximum in two seeds, but the same kneel-and-crawl reference is the recurrent attractor. It passes ordinary kinematic checks. Yet during its 0.75–1.75 s descent, the nearest collision geometry remains 7–10 cm above the floor while a median 329 N of support demand—approximately the 327 N model weight—has no modeled source.

CLIMB closes the interface that produced this ambiguity (Fig. 1). First, *refeas* evaluates the final robot-space trajectory against the target model, scene, contacts, friction, and actuator ranges. Second, DFRP admits supported intervals, restores qualified contact geometry, and quarantines residual or over-distorted repairs. Third, the exact-support curriculum constructs non-wrapping legal starts and applies adaptive learning-progress (ALP) only inside the hard feasibility gate. Policy outcomes close the loop by updating allocation, but never override reference admission.

The resulting evidence covers all three interfaces. The screen flags 2,442/10,705 clips in one AMASS-to-whole\_body\_tracking-to-G1 pipeline and 7/4,950 in a separately filtered BONES-SEED-to-G1 production pairing. The counts are intentionally not pooled: feasibility is a corpus-pipeline property. Two implementations agree on 39/40 strict decisions in a stratified same-clip panel. DFRP qualifies 22/26 flagged panel candidates under residual, displacement, joint-limit, contact-IK, and exact-support gates, while 4/4 feasible controls remain unchanged. Across 100 held-out clips, feasibility features also improve difficulty ranking across policies trained with different curricula from  $\rho = 0.567$  to 0.609 ( $p = 0.010$ ).

This paper makes three contributions:

- 1) the RPM factorization and *refeas*, a final-trajectory screen tested at bank scale and across two implementations;
- 2) DFRP, a fail-closed contact projection that qualifies 22/26 stratified candidates while leaving 4/4 feasible controls unchanged; and
- 3) exact-support ALP: 1,184 units and 368,951 starts, zero rejected mass, and bounded unit/clip allocation.

## II. RELATED WORK

### A. Motion filtering and dynamic retargeting

Policy-based curation already removes poorly tracked references. H2O retains AMASS motions that a privileged imitator can follow, and ExBody2 uses an initial policy’s sequence errors to select a feasible and diverse subset [3], [9]. These scores combine reference quality with policy capability. KungfuBot instead uses a human-space center-of-mass/center-of-pressure heuristic before robot retargeting [10]. LIMMT combines target-robot motion heuristics, diversity, and complexity, with physical-score weights calibrated through repeated policy training [11]. PHUMA filters source artifacts and retargets with joint-limit, ground-contact, and anti-skating losses [12]. CLIMB builds on this line but changes the training interface: the final robot-space screen determines the support on which an adaptive allocator is permitted to act.

Optimization-based retargeters address a complementary problem. Kinodynamic Motion Retargeting uses rigid-body and contact constraints to construct viable locomotion, while Direct Dynamic Retargeting uses simulator-in-the-loop optimization without an infeasible geometric intermediate [13], [14]. AMO and SPIDER likewise generate viable references through trajectory optimization or physics-based sampling with contact guidance [15], [16]. GMR further shows that retargeting artifacts affect downstream tracking robustness [17]. An earlier analytic study estimates velocity, acceleration, and torque requirements for 40 upper-body motions, but does not test whole-body environmental support [18]. Repair can therefore be preferable to exclusion. Our narrower question is diagnostic: after retargeting, can the demanded robot-space wrench be supplied by admissible contacts within modeled actuator limits?

### B. Adaptive allocation and evaluation

Prioritized experience and level replay formalize the benefit and bias of nonuniform training distributions [19], [20]. In humanoid tracking, BeyondMimic uses a failure-EMA bin sampler, while GMT and EGM reweight motions or segments using tracking outcomes [5], [7], [8]. Those systems combine allocation with curation, staging, clipping, or architectural changes. Empirical failure, completion, and tracking error identify hard examples, but do not by themselves determine whether a reference is dynamically admissible. CLIMB inserts that test before outcome-based allocation; its matched comparison then changes only allocation over a shared set of legal trials.

Robot-learning comparisons are sensitive to seed count, aggregation, and interval construction [21]. Tracking adds a conditioning problem: kinematic error among survivors can improve when a method terminates earlier. HumanTracker similarly finds that kinematic metrics can miss support and contact failures [22]. We therefore use a liveness-weighted primary score and report common-survivor quality only beside survival.

## III. CLIMB: DATA-TO-POLICY FRAMEWORK

### A. Model-relative feasibility

Let a retargeted reference provide generalized position and velocity  $(q_t, \dot{q}_t)$  for the G1 at frame  $t$ . We apply a centered five-frame moving average to velocity, differentiate at the reference frame rate, and smooth acceleration with the same window. Feasibility is defined relative to a robot model, scene, collision geometry, friction, actuator ranges, and reference. The present instantiation uses the G1 and a flat plane; electrical, thermal, compliance, latency, and structural limits remain outside this model-relative label.

We use the compiled MuJoCo model underlying the tracking environment. Contact-free inverse dynamics returns the generalized force required for the prescribed acceleration. The six free-joint coordinates define the base wrench  $W_t$  that the environment must supply; the remaining coordinates define contact-free joint torque  $\tau_t^{\text{free}}$ . A collision geometry becomes a candidate contact when its exact distance to the plane is at most  $g = 0.06$  m.

The 6 cm band is a declared tolerance. At 3 cm a known feasible control is flagged on 42% of frames because the bank carries an approximately 3 cm ground-alignment offset. At 10 cm the attractor’s 7–10 cm floating descent is incorrectly granted a contact candidate. The chosen setting lies between these observed failure modes; it is not a universal constant.

### B. *refeas*: contact-capacity screen

For each candidate point we construct a four-edge pyramidal friction cone with coefficient 0.6 in the primary model; a simulator-matched view makes non-foot contacts frictionless. Mapping nonnegative generator magnitudes  $f_t$  through the contact Jacobian yields both base-wrench matrix  $A_t f_t$  and joint-torque contribution  $J_t f_t$ . A weighted nonnegative least-squares problem first exposes the unconstrained residual, with angular-wrench rows weighted by two. We then solve a linear program minimizing  $\ell_1$  wrench slack subject to

$$-\tau_{\max} \leq \tau_t^{\text{free}} - J_t f_t \leq \tau_{\max}, \quad f_t \geq 0. \quad (2)$$

The translational component of the resulting weighted residual defines unsupported force. With no candidate contact, the residual is the contact-free demand.

We report the fraction of frames with no contact candidate (`airborne_frac`), the fraction whose torque-limited unsupported force exceeds half the robot weight (`infeasible_frac`), normalized unsupported-force integral, and joint-torque infeasibility. A clip crosses the primary rule only when `infeasible_frac` > 0.10 (strict inequality). Ballistic flight is not flagged merely for being airborne: near-gravity acceleration has little external support demand. Conversely, nonballistic hovering or descent without support is flagged.

The screen takes approximately one CPU-second per clip in the primary implementation. An independent implementation processes the production bank at 0.145 CPU-s per clip (0.84 ms/frame). The screen therefore runs as an offline bank-ingestion stage rather than in the policy control loop.

### C. DFRP: contact-manifold repair

Dynamic Feasibility and Repair Pipeline (DFRP) targets clips that cross the strict screen rule because collision geometry floats above its intended support surface. Let  $h_t$  be the nearest candidate support geometry,  $d(h_t, E; q_t)$  its signed distance to the scene, and  $c = 3$  mm the target clearance. The repair is the contact-manifold projection

$$\begin{aligned} \min_{\delta z, \Delta \theta} \quad & \sum_{t \in \mathcal{T}} w_z \delta z_t^2 + w_q \|\Delta \theta_t\|_2^2 + w_s (\delta z_{t+1} - \delta z_t)^2 \\ \text{s.t.} \quad & \tilde{q}_t = q_t - \delta z_t e_z + S_t \Delta \theta_t, \\ & g_t = d(h_t, E; \tilde{q}_t) - c \leq 0, \\ & 0 \leq \delta z_t \leq d(h_t, E; q_t) - c, \\ & q_{\min} \leq q_t + S_t \Delta \theta_t \leq q_{\max}. \end{aligned} \quad (3)$$

where  $\mathcal{T}$  contains frames with no collision geometry inside the 6 cm screen band and  $S_t$  selects the six hip, knee, and ankle coordinates of the supporting leg. Equation (3) states the target projection: minimize root and joint deviation while restoring a legal contact without penetrating the plane.

DFRP v1 implements a deterministic two-stage approximation. It first lowers the root by the minimum clearance correction, Gaussian-smooths that timeline over 0.24 s, and clips the result so it never raises the root or pushes the selected geometry through the plane. If the selected leg still misses the contact target, at most 15 damped least-squares iterations apply

$$\Delta \theta_t = \text{clip} \left( -\frac{e_t J_{z,t}^T}{J_{z,t} J_{z,t}^T + 10^{-5}}, -0.05, 0.05 \right), \quad (4)$$

with joint-limit projection after every step. Body poses and velocities are then recomputed by forward kinematics and finite differences. This approximation is fail-closed rather than assumed to satisfy Eq. (3): the full trajectory is re-screened, and admission requires residual infeasibility at most 5%, root displacement at most 8 cm, valid joint limits, maximum contact-IK residual at most 10 mm, and at least one hash-bound legal 50-step start. Scene-mismatch cases, including references that require an absent box, bypass projection and are routed to context completion or quarantine.

### D. Exact-support gated allocation

Clip classification is too coarse because one reference can contain both supportable and unsupported intervals. We reduce per-frame screens to ordered feasible runs, retain only 50-step windows wholly inside a run, and bind each motion and sidecar by SHA-256. Across 800 training clips, 1,841 source runs yield 1,184 admissible units and 368,951 legal starts after 657 short runs are discarded. A unit is an attribution stratum; the deployment prior is uniform over legal starts, not units or clips.

Let  $\mathcal{U}$  contain candidate units,  $F_u \in \{0, 1\}$  denote admission under the declared screen, and  $n_u$  count legal starts. The duration-correct base mass is  $b_u = n_u / \sum_v n_v$ . For absolute

learning progress  $\text{LP}_u(k)$  at sampler clock  $k$ , CLIMB forms

$$\begin{aligned} q_u(k) &= \frac{F_u b_u [\text{LP}_u(k) + \lambda]}{\sum_v F_v b_v [\text{LP}_v(k) + \lambda]}, \\ p(k) &= \text{Cap}_{c_{\text{unit}}, c_{\text{clip}}}(\bar{b}, q(k); \rho), \end{aligned} \quad (5)$$

where  $\bar{b}$  is  $b$  normalized over admitted units. Without an active concentration limit, the operator returns  $\rho \bar{b} + (1 - \rho)q$ ; otherwise it reallocates only the focused mass under the fixed unit and clip ceilings while preserving the exact lower bound  $p_u \geq \rho \bar{b}_u$ . Thus a rejected interval receives zero mass under the declared model, while  $\rho$  preserves deployment-prior exploration inside the gate. The present method uses a binary  $F_u$  so admissibility and allocation remain separately testable; a smooth residual-based gate would be a different intervention.

Intervals do not all receive the same remedy. Admissible runs and DFRP-qualified repairs enter the exact support; missing-context and residual failures do not. This routing preserves feasible subsequences rather than treating every screen flag as a whole-clip deletion instruction.

The runtime samples only legal starts, emits explicit truncation at a segment boundary, never wraps into a rejected frame, and serializes its deterministic generator and sufficient statistics. Per-unit and per-clip caps bound concentration. Missing sidecars, changed motion hashes, invalid starts, invalid frames, or censored resets fail the contract.

The paired evaluator freezes the motion, start, replicate, environment-noise seed, and dynamics-noise seed across policies. It assigns the reference before the first observation, disables auto-reset until terminal channels are captured, rejects invalid offsets, and records checkpoint, task, condition, reference, evaluator, and output hashes. The 100-clip panel is name- and hash-disjoint from training and contains 2,800 paired conditions.

## IV. EXPERIMENTAL DESIGN

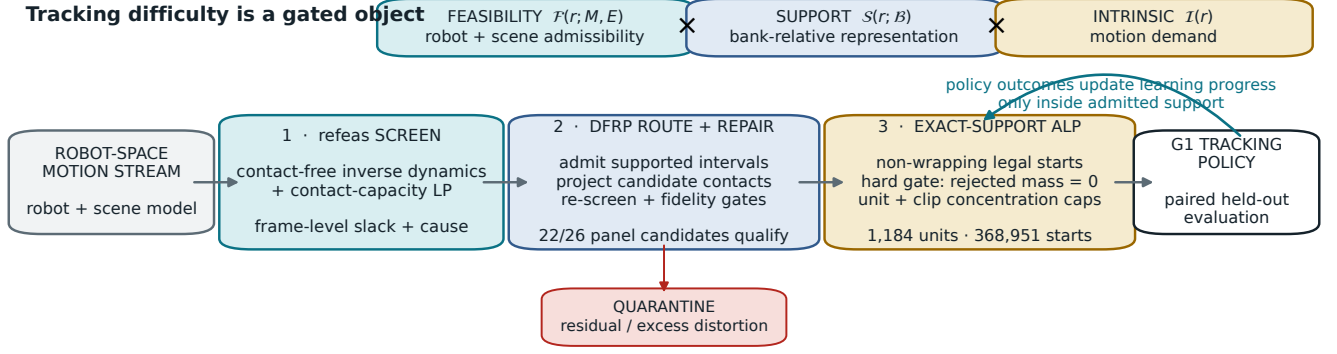
### A. E1: RPM attractor diagnosis

We analyze the previously sealed 100-motion campaign: 4,096 environments, 4,000 PPO iterations, three seeds per arm, and 100 disjoint evaluation motions with eight episodes per clip. It includes failure-adaptive, clip-uniform, and normalized grounded samplers. We report exposure, attractor identity, held-out survival, and the attractor's modeled contact-capacity residual. This campaign establishes the motivating failure case; E4 separately isolates the exact-support allocator.

### B. E2: bank scale, agreement, and transfer

The primary corpus contains 10,705 AMASS-derived references retargeted through `whole_body_tracking` to G1. We report raw counts under the fixed rule, category/source breakdowns, and contamination in the historical evaluator. A separate experiment covers all 4,950 references in the filtered BONES-SEED production pairing. We never pool their prevalence. A deterministic 40-clip panel, 20 from each bank and enriched for flags, passes through both implementations to measure score and decision agreement; it is not a

## CLIMB: close the reference-physics loop before allocating policy updates



Screening changes admission; DFRP recovers qualified data; ALP changes compute allocation. Each interface is measured separately.

Fig. 1. **The CLIMB closed-loop data-to-policy framework.** Difficulty is separated into model-relative feasibility, bank-relative support, and intrinsic demand before policy outcomes affect allocation. *refeas* screens, DFRP repairs and re-qualifies eligible contacts, and exact-support ALP allocates only over legal non-wrapping starts. The 22/26 callout is a frozen stratified repair panel; the 1,184 units and 368,951 starts are properties of the hash-bound 800-motion support, not policy-performance claims.

prevalence sample. Separately, a ridge model fits 11 intrinsic reference features to per-clip difficulty under one curriculum and ranks difficulty under another on the same 100 held-out clips. We compare intrinsic features with and without the three *refeas* features against 200 random three-feature additions; this is cross-policy transfer, not cross-architecture transfer.

### C. E3: DFRP repair qualification

We freeze a source- and severity-stratified CPU panel before repair: 26 strict-flagged candidates span three root-displacement bands and two initial-infeasibility bands, and four source-matched feasible clips serve as no-op controls. The primary outcome is exact-ready flagged count under the 5% residual, 8 cm root, joint-limit, 10 mm contact-IK, source-hash, and legal-start gates. The gate requires at least 20/26 candidates, 4/4 byte-identical ready controls, zero integrity or qualification failures among admitted clips, and a reproducible manifest payload. Runtime and root, joint, body, velocity, and acceleration deviations are secondary diagnostics.

### D. E4: feasibility-gated allocation ablation

Phase G compares two 512-environment, 4,000-iteration arms with confirmation seeds 1–3. G1 samples uniformly over legal starts. G2 estimates each unit’s Beta-smoothed conditional success rate  $s_u(k)$  and ranks absolute learning progress

$$LP_u(k) = |s_u(k) - s_u(k - 10)|. \quad (6)$$

Its focus distribution is proportional to the legal-start prior times  $LP_u + \lambda$ ; exploration  $\rho$  mixes the G1 prior back in before fixed unit and clip caps.

Before sealing, a 12-row grid crosses  $\rho \in \{0.05, 0.10, 0.20, 0.40\}$  and  $\lambda \in \{0.001, 0.01, 0.05\}$ . Each row runs 50 iterations on seed 20260903. Only sampler

ledgers at iterations 30, 40, and 49 enter selection. A row passes when mean TV is in  $[0.05, 0.15]$ , each checkpoint lies in  $[0.025, 0.20]$ , entropy-effective units are at least 12, top-1 probability is at most 0.05, invalid/censored counts are zero, and final rank saturation is below 0.90. The passing row nearest TV 0.10 is repeated on seed 20260904. Failure returns the study to design without endpoint access.

The endpoint-blind calibration selected  $\rho = 0.40$ ,  $\lambda = 0.05$ . Its screen TVs are 0.1310/0.1063/0.0865 (mean 0.1079); the independent-seed TVs are 0.1292/0.1045/0.0831 (mean 0.1056). The latter retains at least 700.1 entropy-effective units, at most 0.0134 top-1 mass, zero invalid or censored events, and 0.2365 final saturation. This is a passed manipulation calibration, not an ALP policy-performance result; the G2–G1 endpoint remains sealed and unread.

After a 400-iteration warm-up, the confirmation gate applies the same separation, concentration, validity, seed, and provenance checks. Only a passed gate unlocks the primary endpoint: liveness-weighted TrackingScore on 25 reference-defined feasible-hard clips,

$$h \exp\left(-\frac{\text{MPKPE}}{0.30 \text{ m}} - \frac{\text{anchor error}}{0.40 \text{ rad}}\right), \quad (7)$$

where  $h$  is survived-horizon fraction. The independent unit is clip within training seed, with a seed-then-clip hierarchical bootstrap of 10,000 draws. Survival, all-panel TrackingScore, AULC, and common-survivor non-harm are secondaries. Exactly one status is printed: positive, null, inconclusive, or `not_tested`.

## V. RESULTS

### A. RPM creates a curriculum attractor

Adaptive peak top-1 mass is 0.884/0.870/0.893 across campaign seeds, and the same BMLmovi kneel-and-crawl clip repeatedly becomes dominant in all three. The maximum

belongs to another clip in two seeds. Uniform sampling achieves higher held-out endpoint survival in every paired seed: differences (uniform minus adaptive) are +0.0300, +0.0275, and +0.0300. With only three seeds, the sign pattern is the useful description; the minimum attainable paired permutation  $p$ -value is 0.125.

The reference has no joint-limit violation and peak joint speed is at most 5.6 rad/s. The dynamic screen instead localizes the failure. From 0.75–1.75 s, 86% of frames have no candidate contact and median unsupported force is 329 N; the model weighs 327 N. The 1.75–7.25 s kneeling interval is supportable through non-foot contacts, and the 8.0–8.5 s rise again becomes unsupported. Every trained policy has zero survival from frame zero, while the tested late 8 s offset survives.

### B. *refeas* is pipeline-conditioned and reproducible

Under the strict rule, 2,442 of 10,705 primary-pipeline clips are flagged (22.8%). Category rates range from 12.9% for quiet motion to 58.6% for dynamic motion, and source rates from 0.1% for GRAB to 100% for CNRS. Severity-stratified inspection shows why labels are insufficient: five inspected CNRS cases are ordinary fast walks with extended floating intervals, while the Transitions sample mixes ingest defects with acrobatic content.

Only 7 of 4,950 production clips (0.14%) cross the same nominal rule, while 111 (2.24%) are airborne on more than 10% of frames. Five of seven flags are jumps, four naming a 50 cm box absent from the declared flat scene. Their verdict is *unsupported on this plane*, routing them toward missing terrain or exclusion rather than geometric root projection.

Across the stratified 40-clip panel, `infeasible_frac` rank correlation is 0.9836, `airborne_frac` correlation is 0.9974, and strict decisions agree for 39/40 clips (Cohen's  $\kappa = 0.9485$ ). The disagreement is `burpee_002__A362_M` (0.019 versus 0.136). Figure 2 keeps the full-bank denominators separate and uses only this same-input panel to check implementation agreement; it does not establish corpus or retargeter equivalence.

The difficulty decomposition is also visible across policies trained with different curriculum allocations. On the same 100 held-out clips, an 11-feature intrinsic atlas fit to adaptive-policy difficulty ranks uniform-policy difficulty at  $\rho = 0.567$ . Adding `infeasible_frac`, `airborne_frac`, and normalized unsupported impulse raises the rank correlation to 0.609, above 198 of 200 random three-feature additions (one-sided  $p = 0.010$ ). All six directed policy pairs move in the same direction and four have  $p < 0.05$ ; the two pairs targeting the grounded policy do not. This is transfer across policies sharing an architecture, not evidence of cross-architecture or cross-robot transfer.

### C. DFRP restores exact support on a stratified panel

DFRP qualifies 22/26 flagged candidates (84.6%) and all four feasible controls pass as byte-identical no-ops. The resulting 26-clip curated view contains 36 admissible units and 10,561 legal 50-step starts. Median and 95th-percentile

CPU runtime are 2.57 and 6.29 s per clip. Two candidates remain above the 5% residual-infeasibility ceiling, and two more reach that ceiling but violate the 10 mm contact-IK bound; all four are quarantined. Thus 84.6% is a qualification rate on the frozen panel, not a bank-wide recovery rate or a policy-benefit estimate.

### D. Alternative routing and allocation controls

The historical evaluator contains 29 flagged clips among 100. Across three policies, flagged clips score 6.0–8.4 survival points below the all-clip aggregate and 8.4–11.8 below the feasible stratum. The grounded-minus-uniform endpoint contrast is +0.025 on feasible clips and –0.009 on flagged clips. These data justify stratified evaluation, but changing the evaluation subset cannot establish training benefit.

These controls explain why each CLIMB interface is separate. Whole-clip E-HYG pruning removes all 99 flagged motions even though exact segmentation retains 12.5 of their 20.2 minutes and loses only 3/99 clips at zero guard; its feasible-held-out survival changes from 0.918 to 0.907 ( $\Delta = -0.0101$ , one-sided permutation  $p = 0.951$ ). The tested soft segment weighting likewise changes feasible-hard survival by –0.0196 with 95% hierarchical-bootstrap interval [–0.0497, +0.0134] and leaves 0.199 mass on hard-rejected starts, failing its predeclared manipulation gate. These outcomes do not establish a universal failure of pruning or soft weighting; they show that neither tested substitute implements exact admission.

An earlier exact-support pilot provides a bounded fidelity signal rather than an allocation win. Across 22,321 paired common-survivor frames, the adaptive-minus-uniform body-position error is –4.20 mm (95% unit-bootstrap interval [–6.63, –1.90] mm) and anchor-orientation error is –0.02795 rad ([–0.03966, –0.01628] rad). The allocation changes by only 0.014 TV, however, so survival and success remain inconclusive and the pilot cannot attribute those improvements to ALP. Similarly, an earlier repair-all policy study estimates a +0.0397 deployment change but misses its +0.05 smallest effect of interest and its coverage gate. Negative controls therefore remain design evidence: they separate detection, admission, repair qualification, and policy allocation instead of treating them as one intervention.

## VI. LIMITATIONS

CLIMB's admission label is model-relative. The current screen instantiates one G1 model and a flat scene with collision contacts, friction, and actuator force limits. Hardware deployment would add electrical, thermal, compliance, latency, and structural constraints and then require closed-loop validation. Terrain-bearing references likewise require their terrain in the scene; the four box-jump flags in the production bank are routed to missing context rather than declared bad motion.

Finite differences, the 6 cm contact band, and the half-weight residual threshold introduce modeling choices. Local sensitivity checks bound the two thresholds on this bank, but a new robot or retargeter must recalibrate them. The 22.8%

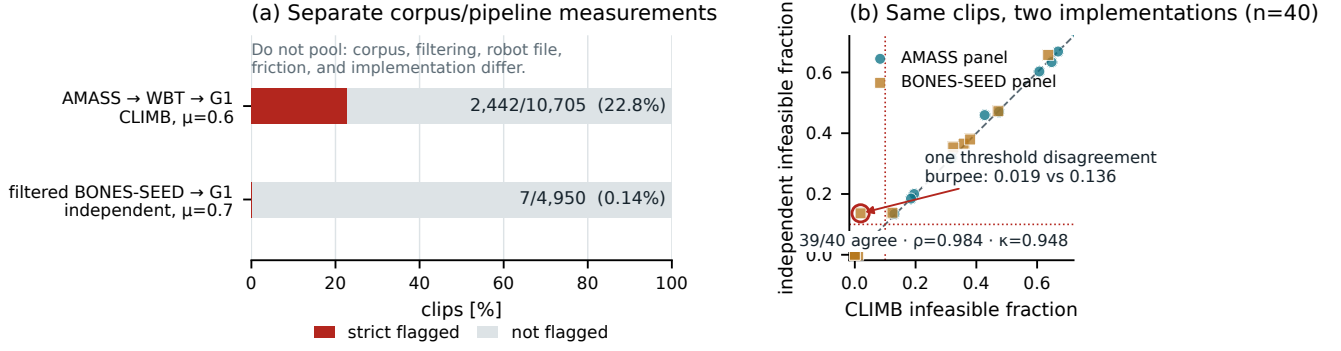


Fig. 2. Bank-scale screen and implementation agreement. (a) Bars are normalized within two separate corpus/pipeline denominators and apply the strict `infeasible_frac > 0.10` rule: 2,442/10,705 for AMASS→WBT→G1 and 7/4,950 for filtered BONES-SEED→G1. Corpus, filtering, robot file, friction, and implementation differ, so this is not a causal retargeter comparison. (b) On a flag-enriched, deterministic 20+20 same-clip panel, strict decisions agree for 39/40 clips (Spearman  $\rho = 0.9836$ , Cohen’s  $\kappa = 0.9485$ ). That panel checks implementation agreement; its stratified selection is not a prevalence sample.

and 0.14% rates therefore remain separate corpus–pipeline measurements. A smooth feasibility weight derived from residual slack could retain borderline cases, but would couple admissibility and allocation; it requires its own matched ablation against the current binary gate.

The exact-support ALP sampler and its matched multi-seed protocol are implemented, but this manuscript does not report a passed ALP-versus-uniform endpoint. Zero rejected mass and the unit and clip caps are therefore mechanism properties, not evidence of policy superiority. The repair route likewise needs a same-policy, distortion-aware comparison of certified repair against exact feasible intervals and quarantine. The observation that would settle this is joint movement of tracking, survival, reference fidelity, contact timing, and mechanical work under one paired trial contract.

## VII. CONCLUSION

Reference–physics misalignment turns persistent tracking error into a misleading curriculum signal. In the diagnosed three-seed campaign, one such reference repeatedly becomes a dominant attractor; `refeas` localizes its unsupported wrench demand before policy training. CLIMB turns that diagnosis into an active interface: DFRP restores qualified contacts, exact-support sampling excludes rejected frames, and policy outcomes allocate compute only inside that gate. On the tested artifacts, DFRP qualifies 22/26 stratified repair candidates, the hard support exposes 368,951 legal starts, and feasibility features improve difficulty transfer across policies from  $\rho = 0.567$  to 0.609. Feasibility, data support, intrinsic demand, and learning progress are therefore separate quantities that a humanoid tracking system can measure and change without asking one noisy failure signal to represent all four.

## REPRODUCIBILITY AND AI DISCLOSURE

The screening and training stack uses `mjlab v1.6.0` at commit `0fb8a681136b`, `MuJoCo 3.11.0`, `NumPy 2.5.1`, and `SciPy 1.16.2`. Motion, sidecar, unit-table, task, checkpoint, evaluator, condition, and output identities are SHA-256

bound. Released code and aggregate artifacts will be supplied through an anonymous review repository; licensed AMASS-derived motion files cannot be redistributed. OpenAI Codex assisted with code, figure composition, and language editing. The authors verified scientific claims and artifact provenance.

## REFERENCES

- [1] Z. Luo, J. Cao, A. Winkler, K. Kitani, and W. Xu, “Perpetual humanoid control for real-time simulated avatars,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2023, pp. 10 895–10 904.
- [2] C. Tessler, Y. Guo, O. Nabati, G. Chechik, and X. B. Peng, “Masked-Mimic: Unified physics-based character control through masked motion inpainting,” *ACM Transactions on Graphics*, vol. 43, no. 6, pp. 209:1–209:21, 2024.
- [3] T. He, Z. Luo, W. Xiao, C. Zhang, K. Kitani, C. Liu, and G. Shi, “Learning human-to-humanoid real-time whole-body teleoperation,” 2024, arXiv:2403.04436.
- [4] X. Cheng, Y. Ji, J. Chen, R. Yang, G. Yang, and X. Wang, “Expressive whole-body control for humanoid robots,” in *Robotics: Science and Systems*, 2024.
- [5] Q. Liao, T. E. Truong, X. Huang, Y. Gao, G. Tevet, K. Sreenath, and C. K. Liu, “BeyondMimic: From motion tracking to versatile humanoid control via guided diffusion,” 2025, arXiv:2508.08241.
- [6] Z. Luo, Y. Yuan, T. Wang, C. Li, F. Castañeda, S. Chen, Z.-A. Cao, J. Li, D. Minor, Q. Ben, J. Park, D. Sami, Z. Wang, X. Da, R. Ding, C. Hogg, L. Song, E. Lim, E. Jeong, T. He, H. Xue, W. Xiao, S. Yuen, J. Kautz, Y. Chang, U. Iqbal, L. J. Fan, and Y. Zhu, “SONIC: Supersizing motion tracking for natural humanoid whole-body control,” 2025, arXiv:2511.07820.
- [7] Z. Chen, M. Ji, X. Cheng, X. Peng, X. B. Peng, and X. Wang, “GMT: General motion tracking for humanoid whole-body control,” 2025, arXiv:2506.14770.
- [8] C. Yang, Y. Sun, P. Ye, X. Chen, C. Yu, and T. Chen, “EGM: Efficiently learning general motion tracking policy for high dynamic humanoid whole-body control,” 2025, arXiv:2512.19043.
- [9] M. Ji, X. Peng, F. Liu, J. Li, G. Yang, X. Cheng, and X. Wang, “Ex-Body2: Advanced expressive humanoid whole-body control,” 2024, arXiv:2412.13196.
- [10] W. Xie, J. Han, J. Zheng, H. Li, X. Liu, J. Shi, W. Zhang, C. Bai, and X. Li, “KungfuBot: Physics-based humanoid whole-body control for learning highly-dynamic skills,” in *Advances in Neural Information Processing Systems*, vol. 38, 2025.
- [11] Y. Guan, Z. Qi, C. Lin, X. Chen, D. Liu, W. Zhang, J. Wang, X. Yu, H. Wang, and L. Yi, “LIMMT: Less is more for motion tracking,” in *International Conference on Machine Learning*, 2026.
- [12] K. Lee, S. Kim, M. Park, H. Kim, D. Hwang, H. Lee, and J. Choo, “PHUMA: Physically-grounded humanoid locomotion dataset,” 2025, arXiv:2510.26236.

- [13] X. Zhang, S. Haener, V. Madabushi, and M. Tucker, “Kinodynamic motion retargeting for humanoid locomotion via multi-contact whole-body trajectory optimization,” 2026, arXiv:2603.09956.
- [14] C. Roux, L. D. Matteis, A. Jordana, V. Guillet, N. Mansard, O. Stasse, and P. Souères, “Direct dynamic retargeting for humanoid imitation learning from videos,” 2026, arXiv:2605.23762.
- [15] J. Li, X. Cheng, T. Huang, S. Yang, R.-Z. Qiu, and X. Wang, “AMO: Adaptive motion optimization for hyper-dexterous humanoid whole-body control,” in *Robotics: Science and Systems*, 2025.
- [16] C. Pan, C. Wang, H. Qi, Z. Liu, H. Bharadhwaj, A. Sharma, T. Wu, G. Shi, J. Malik, and F. Hogan, “SPIDER: Scalable physics-informed dexterous retargeting,” 2025, arXiv:2511.09484.
- [17] J. P. Araujo, Y. Ze, P. Xu, J. Wu, and C. K. Liu, “Retargeting matters: General motion retargeting for humanoid motion tracking,” 2025, arXiv:2510.02252.
- [18] C. Klas, A. Meixner, D. Ruffler, and T. Asfour, “On the actuator requirements for human-like execution of retargeted human motion on humanoid robots,” in *2023 IEEE-RAS 22nd International Conference on Humanoid Robots*, 2023, pp. 1–8.
- [19] T. Schaul, J. Quan, I. Antonoglou, and D. Silver, “Prioritized experience replay,” in *International Conference on Learning Representations*, 2016.
- [20] M. Jiang, E. Grefenstette, and T. Rocktäschel, “Prioritized level replay,” in *Proceedings of the 38th International Conference on Machine Learning*, 2021, pp. 4940–4950.
- [21] R. Agarwal, M. Schwarzer, P. S. Castro, A. Courville, and M. G. Bellemare, “Deep reinforcement learning at the edge of the statistical precipice,” in *Advances in Neural Information Processing Systems*, vol. 34, 2021, pp. 29 304–29 320.
- [22] D. Liu, Z. Qi, J. Zeng, R. Yu, Y. Guan, Y. Zhang, X. Chen, S. Liang, Z. Li, C. Lin, X. Yu, W. Zhang, H. Wang, and L. Yi, “HumanTracker: Towards comprehensive and human-aligned motion tracking benchmark,” in *European Conference on Computer Vision*, 2026.